Level 3 / Practice and review

16 subject tests. Questions first, explained answer keys afterward. Use a mistake log and retry missed concepts after a delay.

Mechanics and Heat / Test A

25 marks | Suggested time: 25 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. A runner completes one lap and returns to the start. Average velocity is:

   A. Always positive

   B. Impossible to define

   C. Zero

   D. Equal to average speed

2. The area under a velocity-time graph gives:

   A. Displacement

   B. Force

   C. Speed at the final instant

   D. Acceleration

3. Two perpendicular forces of 6 N and 8 N have resultant magnitude:

   A. 10 N

   B. 48 N

   C. 2 N

   D. 14 N

4. At the top of a projectile path, acceleration is:

   A. Zero

   B. Horizontal

   C. Upward

   D. Downward with magnitude g

5. An object moving at constant velocity has:

   A. A forward net force

   B. Increasing acceleration

   C. Zero net force

   D. No forces acting at all

Part II: Show your reasoning

1. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

2. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

3. Find the magnitude and direction of a vector with components -3 m and +4 m.

4. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use $g = 9.8\ m/s^2$. Find flight time and horizontal range.

5. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use $g = 10\ m/s^2$. Find normal force, friction and acceleration.

Mechanics and Heat / Test B

25 marks | Suggested time: 25 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Doubling speed at fixed radius changes centripetal acceleration by a factor of:

   A. Two

   B. Four

   C. Eight

   D. One half

2. A force perpendicular to displacement does work equal to:

   A. Fd

   B. Zero

   C. -Fd

   D. Half of Fd

3. If the extension of a Hooke-law spring doubles while k stays constant, its stored energy becomes:

   A. Four times as large

   B. Twice as large

   C. Unchanged

   D. Half as large

4. During melting at fixed pressure, added heat mainly:

   A. Raises temperature continuously

   B. Changes phase

   C. Decreases mass

   D. Disappears

5. Which heat-transfer mechanism can operate through a vacuum?

   A. Fluid mixing

   B. Convection

   C. Conduction through air

   D. Radiation

Part II: Show your reasoning

1. An ideal Atwood machine has masses 2 kg and 3 kg. Use g = 10 m/s^2. Find acceleration and tension.

2. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

3. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

4. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

5. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Electricity and Magnetism / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. A neutral object loses electrons. It becomes:

   A. Negative

   B. Positive

   C. Uncharged forever

   D. Magnetic only

2. A negative test charge in a field directed right feels force:

   A. Left

   B. Zero

   C. Right

   D. Up

3. A charge outside a closed empty box gives net flux through the box of:

   A. $\frac{q}{\varepsilon_0}$

   B. Always positive

   C. Always negative

   D. Zero

4. Inside the material of a conductor in electrostatic equilibrium, the electric field is:

   A. Equal to its surface value

   B. Infinite

   C. Zero

   D. Always outward

5. For equal and opposite point charges, take potential as zero at infinity. At their midpoint the potential is:

   A. Zero

   B. Undefined

   C. Positive

   D. Negative

6. Adding a dielectric to an isolated charged capacitor increases C. Its voltage:

   A. Decreases

   B. Increases

C. Must stay fixed

D. Becomes infinite

Part II: Show your reasoning

1. Two charges exert 12 N at separation r. What is the force at separation 2r, and at r if one charge is tripled?

2. A field of 200 N/C points east. Find the force on a charge of -0.003 C.

3. A uniform field of 50 N/C passes through a flat area of 0.20 square metres. The chosen surface normal points in the same direction as the field. Find flux. What if the normal is perpendicular to the field?

4. An isolated charged conducting sphere, with no external fields, has field magnitude 80 N/C at r = 0.20 m. Find the field at r = 0.40 m. Both distances are from the centre and outside the sphere.

5. A +2 C charge moves through a potential difference of +3 V. Find its potential-energy change and the work done by the field.

6. A 4 μF capacitor is connected to 12 V. Find the charge in microcoulombs.

Electricity and Magnetism / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. At fixed voltage, doubling resistance makes power:

   A. Unchanged

   B. Double

   C. Half

   D. Four times

2. After a long time in an ideal DC charging circuit, capacitor current is:

   A. Maximum

   B. Negative forever

   C. Zero

   D. Equal to the battery voltage divided by the resistor resistance

3. A charge moving parallel to a magnetic field experiences magnetic force:

   A. qvB

   B. Twice qvB

   C. Opposite its velocity

   D. Zero

4. Two parallel wires carrying currents in the same direction:

   A. Repel

   B. Attract

   C. Have no interaction

   D. Must carry zero field

5. Which situation produces an induced emf?

   A. No change of area, field or angle

   B. Changing magnetic flux

   C. A stationary unchanging field only

   D. Constant flux

6. In vacuum, electric and magnetic fields of a plane EM wave are:

   A. Parallel to each other

   B. Longitudinal only

C. Zero

D. Perpendicular to each other and to travel

Part II: Show your reasoning

1. A 3 Ω resistor carries 2 A. Find voltage and power.

2. A 4 Ω and 8 Ω resistor are in series across 24 V. Find total current and each voltage drop.

3. A straight 0.50 m wire carries 2 A perpendicular to a 0.30 T field. Find force magnitude.

4. The field is 30 microtesla at 2 cm from a long straight wire. Find the field at 6 cm at the same current.

5. A 20-turn coil has flux per turn changing by 0.03 Wb in 0.10 s. Find average emf magnitude.

6. A sinusoidal voltage has peak value 20 V. Find its RMS value, then the average power in a 10 Ω resistor.

Optics and Waves / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which wave can propagate through vacuum?

   A. A water wave

   B. Light

   C. A stretched-string wave

   D. Sound in air

2. With wave speed fixed, doubling frequency makes wavelength:

   A. Half as large

   B. Unchanged

   C. Four times as large

   D. Twice as large

3. Group velocity describes the motion of:

   A. The wave-packet envelope

   B. The source itself

   C. One particle only

   D. A stationary node

4. Two tones at 440 and 443 Hz produce a beat frequency of:

   A. 1320 Hz

   B. 883 Hz

   C. 3 Hz

   D. 441.5 Hz

5. At the same medium, which has greater photon energy?

   A. Lower-frequency light

   B. Both always equal

   C. Higher-frequency light

   D. Longer-wavelength light

6. Light enters glass from air. Which quantity remains unchanged?

   A. Ray direction in every case

   B. Speed

C. Wavelength

D. Frequency

Part II: Show your reasoning

1. A source makes 50 complete oscillations in 10 s. Find frequency and period.

2. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

3. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

4. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

5. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

6. Find the critical angle for glass of index 1.5 into air of index 1.0.

Optics and Waves / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. An image with magnification -2 is:

   A. Upright and smaller

   B. Inverted and twice as tall

   C. Virtual in every case

   D. Upright and twice as tall

2. In a simple astronomical telescope, the objective usually has:

   A. A longer focal length than the eyepiece

   B. Zero focal length

   C. The same role as a polarizer

   D. No effect on the image

3. If double-slit separation increases while wavelength and screen distance stay fixed, fringe spacing:

   A. Becomes negative

   B. Increases

   C. Stays fixed

   D. Decreases

4. Two ideal crossed polarizers transmit from an initially unpolarized beam:

   A. One quarter

   B. Zero

   C. All its intensity

   D. Half

5. The unit of optical lens power is:

   A. Hertz

   B. Watt

   C. Dioptre

   D. Newton

6. A quarter-wave plate acts primarily by changing the relative:

   A. Charge of light

   B. Speed of the source

C. Mass of photons

D. Phase of orthogonal components

Part II: Show your reasoning

1. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

2. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

3. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

4. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

5. A diverging lens has focal length -50 cm. Find its optical power.

6. For light from air into glass of index 1.5, estimate Brewster angle.

Calculus I / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. The derivative of x^2 at $x = 3$ is:

   A. 3

   B. 6

   C. 9

   D. 0

2. The derivative of a constant 7 is:

   A. 7

   B. 1

   C. x

   D. 0

3. The derivative of xe^x is:

   A. $e^x(1 + x)$

   B. 0

   C. e^x

   D. xe^x

4. The derivative of $\sin(5x)$ is:

   A. $-5\sin(5x)$

   B. $\sin(5)$

   C. $\cos(5x)$

   D. $5\cos(5x)$

5. For $x^2 + y^2 = 25$, the slope at (3,4) is:

   A. 0

   B. $\frac{3}{4}$

   C. $-\frac{3}{4}$

   D. $\frac{4}{3}$

6. The derivative of $\arctan x$ is:

   A. $\arctan(1)$

   B. $\frac{1}{1+x^2}$

C. $\frac{1}{\cos^2 x}$

D. $-\frac{1}{1+x^2}$

Part II: Show your reasoning

1. Use the definition to find the derivative of f(x)=$3x^2$.

2. Find the tangent to f(x)=x cubed at $x = 2$.

3. Differentiate y=(x+1)/(x-1), for x not equal to 1.

4. Differentiate y=cos(x^2).

5. Find dy/dx for xy+y squared=6.

6. Differentiate y=x to the power x for x greater than zero.

Calculus I / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Before applying L Hopital directly, the quotient should have form:

   A. $\frac{0}{0}$ or $\frac{\infty}{\infty}$

   B. $\frac{1}{1}$ only

   C. $\frac{0}{5}$ only

   D. Any fraction

2. If f' is negative on an interval, f is:

   A. Decreasing

   B. Constant

   C. Always concave down

   D. Increasing

3. To find absolute extrema on a closed interval, check:

   A. Only where f' is zero

   B. Only endpoints

   C. Endpoints and interior critical numbers

   D. Only where f is zero

4. Rolle theorem additionally requires:

   A. An infinite interval

   B. Equal endpoint function values

   C. A discontinuity

   D. A nonzero derivative everywhere

5. An indefinite integral must generally include:

   A. Only a numerical answer

   B. A fixed upper limit

   C. No variable

   D. An arbitrary constant

6. The $\int_{-2}^{2} x\,dx$ equals:

   A. 8

   B. -4

C. 0

D. 4

Part II: Show your reasoning

1. Evaluate the limit of sin(3x)/x as x approaches zero.

2. For $f(x) = x^2$-4x, find increasing and decreasing intervals and the minimum point.

3. Find absolute maximum and minimum of $f(x) = x^2$-2x on [0,3].

4. Use the mean-value theorem for $f(x) = x^2$ on [1,3] to find c.

5. Evaluate the integral of 6x(x^2+2) squared with respect to x.

6. Evaluate the integral from 0 to 1 of 2x(x^2+1) dx.

Mechanics and Heat Lab / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. If y is force in N and x is extension in m, slope units are:

   A. N/m

   B. m/N

   C. N times m

   D. No units

2. Pendulum length should be measured to:

   A. The top of the table

   B. The lowest swing point

   C. The bottom of the bob

   D. The centre of the bob

3. For t^2 vertically versus h horizontally, g equals:

   A. Slope squared

   B. Slope/2

   C. 2 times slope

   D. 2/slope

4. The spring constant is the slope of:

   A. Time versus temperature

   B. Extension versus force

   C. Force versus extension

   D. Mass versus time

5. For fixed k, multiplying the oscillating mass by four makes the period:

   A. Four times

   B. Half

   C. Twice

   D. Unchanged

6. The equilibrant of a 7 N eastward resultant is:

   A. 7 N east

   B. 14 N west

C. Zero

D. 7 N west

Part II: Show your reasoning

1. A measured g is 9.6 m/s^2 and reference g is 9.8 m/s^2. Calculate percentage error and name one possible systematic error.

2. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

3. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

4. A 0.30 kg mass extends a spring by 6 cm. Use g=10 m/s^2. Find k.

5. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

6. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Mechanics and Heat Lab / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. In the simple cooling model, the cooling rate is proportional to:

   A. Mass only

   B. Zero

   C. Temperature excess above surroundings

   D. Absolute time only

2. Boyle law requires:

   A. Fixed temperature and gas amount

   B. Changing gas amount

   C. Fixed pressure

   D. Fixed volume

3. Two rods are made of the same material and have the same temperature rise. A rod twice as long has expansion:

   A. Twice as large

   B. Unchanged

   C. Four times as large

   D. Half as large

4. Electrical energy supplied by constant voltage and current for time t is:

   A. V/I

   B. IVt

   C. I/t

   D. Vt/I

5. In an ideal mixing experiment, the final equilibrium temperature lies:

   A. Below both

   B. Between the initial temperatures

   C. Always at zero

   D. Above both initial temperatures

6. To determine joules per calorie, divide electrical energy in joules by heat measured in:

   A. Watts

   B. Calories

C. Kelvin

D. Joules

Part II: Show your reasoning

1. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

2. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

3. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

4. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

5. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

6. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Instruments and Lab Safety / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which best describes risk?

   A. The mass of any bottle

   B. The same as smell

   C. The label colour alone

   D. The chance and severity of harm under the exposure conditions

2. Which is appropriate before touching a clean door handle?

   A. Keep contaminated gloves on

   B. Wipe gloves on clothing

   C. Remove contaminated gloves correctly

   D. Smell the gloves

3. Eating with contaminated hands is mainly which exposure route?

   A. Inhalation

   B. Injection

   C. Noise exposure

   D. Ingestion

4. An oxidizer is especially associated with:

   A. Intensifying combustion

   B. Preventing all fires

   C. Measuring temperature

   D. Making all chemicals inert

Part II: Show your reasoning

1. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

2. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

3. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

4. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Instruments and Lab Safety / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is designed to prepare one accurately specified solution volume?

   A. A beaker

   B. A watch glass

   C. A volumetric flask

   D. A test-tube rack

2. A fume hood should be used as:

   A. Permanent chemical storage

   B. A controlled work enclosure for appropriate vapour hazards

   C. A way to evaporate all waste

   D. A food cupboard

3. An unknown chemical spill should first prompt a student to:

   A. Alert the instructor and keep people clear

   B. Mix it with another chemical

   C. Ignore it

   D. Taste it

4. Which is an administrative control?

   A. Gloves

   B. Training and written procedures

   C. A sealed rotor

   D. Splash goggles

Part II: Show your reasoning

1. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

2. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

3. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

4. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Humans and the Environment / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is an abiotic factor?

   A. Algae

   B. Bacteria

   C. Water temperature

   D. Fish

2. Decomposers mainly help by:

   A. Recycling nutrients from dead matter

   B. Creating sunlight

   C. Stopping all energy loss

   D. Removing all producers

3. Which directly reduces a local population?

   A. Reproduction

   B. Births

   C. Immigration

   D. Emigration

4. Both organisms benefit in:

   A. Parasitism

   B. Mutualism

   C. Predation

   D. Competition

Part II: Show your reasoning

1. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

2. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

3. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

4. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Humans and the Environment / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Excess nutrient input into water is most closely associated with:

   A. Eutrophication

   B. Lens refraction

   C. Sterilization

   D. No biological change

2. A footprint larger than biocapacity indicates:

   A. Unlimited sustainability

   B. An ecological reserve

   C. An ecological deficit

   D. Zero resource use

3. Which is highest in the general waste hierarchy?

   A. Disposal after mixing

   B. Landfill

   C. Uncontrolled burning

   D. Waste prevention

4. A complete sustainability assessment considers:

   A. Only this month

   B. Environmental, social and economic effects

   C. Only profit

   D. Only tree numbers

Part II: Show your reasoning

1. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

2. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

3. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

4. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Professional Development Seminar / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is a professional criterion emphasized in the lecture?

   A. No responsibility to society

   B. Learning stops at graduation

   C. No preparation

   D. Specialized knowledge and ethical standards

2. Using student work to improve instruction best illustrates the teacher as:

   A. A storage clerk only

   B. An assessor and learning guide

   C. A person who never changes plans

   D. A passive spectator

3. Support during a teacher first years of practice is mainly:

   A. Only pre-service admission

   B. Induction

   C. Retirement

   D. Waste management

4. The central value of supervised field experience is:

   A. Eliminating evaluation

   B. Replacing all theory

   C. Connecting theory to practice with feedback

   D. Avoiding learners

Part II: Show your reasoning

1. Give two reasons teaching is considered a profession, each with an example.

2. Describe two teacher roles beyond explaining subject content.

3. Compare pre-service preparation with induction in two clear points.

4. Give a four-step improvement cycle for a trainee after an observed lesson.

Professional Development Seminar / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is the strongest development goal?

   A. Improve one identified practice and evaluate its effect

   B. Collect the most certificates

   C. Become better somehow

   D. Attend anything

2. What makes a portfolio stronger?

   A. No evaluation

   B. More unrelated documents

   C. Evidence linked to goals and reflection

   D. Only decorative pages

3. Constructive professional feedback should be:

   A. Only praise regardless of evidence

   B. Unrelated to teaching

   C. Personal and humiliating

   D. Based on observation and useful next steps

4. The best reason to choose an educational tool is:

   A. It supports a defined learning objective

   B. It has the most animations

   C. Everyone must use the same tool

   D. It is newest

Part II: Show your reasoning

1. Write a short development plan for a teacher whose students struggle to explain answers.

2. Name two portfolio items and explain what each can demonstrate.

3. Rewrite this feedback constructively: Your lesson was bad.

4. Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

Mechanics and Heat / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** The net displacement is zero even though the distance travelled is not.
2. **A.** Area has units (m/s) times s = m and includes the sign of velocity.
3. **A.** The Pythagorean calculation gives the square root of 36+64 = 10 N.
4. **D.** Gravity continues to act at the top.
5. **C.** Constant velocity means zero acceleration, so forces balance.

Part II: Written solutions

1. Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity.
2. a = (8-20)/4 = -3 m/s^2. s = [(20+8)/2] times 4 = 56 m. Two marks for each calculation, including the sign and unit.
3. Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction.
4. 4.9 = 0.5 times 9.8 times t^2, so t = 1 s. Range = 5 times 1 = 5 m. Two marks per result with the equation shown.
5. N = 50 N, friction = 0.2 times 50 = 10 N, net force = 10 N, a = $1\frac{0}{5}$ = 2 m/s^2. One mark for each step.

Mechanics and Heat / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The speed is squared.
2. **B.** The cosine of 90 degrees is zero.
3. **A.** Spring energy is proportional to extension squared.
4. **B.** Energy changes the phase while temperature remains at the melting point.
5. **D.** Electromagnetic radiation needs no material medium.

Part II: Written solutions

1. Net driving force on the two-mass system is (3-2) times 10 = 10 N. Total mass = 5 kg, so a = 2 m/s^2. For the rising 2 kg mass, T-20 = 2 times 2, hence T = 24 N. Two marks per result.
2. Net work = 0.5 times 2 times (25-9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit.
3. Spring energy = 0.5 times 200 times 0.10 squared = 1 J. Set 1 = 0.5 times 0.50 times v squared to get v = 2 m/s. Two marks for each stage.
4. Temperature rise = 10 K. Q = mc times temperature rise = 0.50 times 4200 times 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit.
5. Increase = 0.000012 times 2 times 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion.

Electricity and Magnetism / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** Removing negative charge leaves a positive net charge.
2. **A.** The negative charge reverses the field direction in $F = qE$.
3. **D.** No charge is enclosed, so total entering and leaving flux cancels.
4. **C.** Free charges redistribute until the internal field is cancelled.
5. **A.** The signed scalar potentials cancel when distances and charge magnitudes match.
6. **A.** Charge remains fixed and $V = \frac{Q}{C}$.

Part II: Written solutions

1. At 2r: 12/4 = 3 N. Tripling one charge at r: 36 N. Two marks for each ratio and result.
2. Magnitude = 0.003 times 200 = 0.60 N. Direction is west because the charge is negative. Two marks for magnitude and two for direction with reasoning.
3. Parallel normal: flux = 50 times 0.20 = 10 N m squared/C. Perpendicular normal: cosine 90 degrees = 0, so flux = 0. Two marks per case.
4. Inverse-square scaling gives 80 times $(0.2\frac{0}{0}.40)$ squared = 20 N/C. Credit the outside-sphere condition and inverse-square reasoning.
5. Energy change = 2 times 3 = +6 J. Work by field = -6 J. Two marks each; the signs are essential.
6. Q = CV = 4 times 12 = 48 microcoulombs. In SI this is 0.000048 C. Credit formula, substitution, prefix and result.

Electricity and Magnetism / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** $P = \frac{V^2}{R}$, so doubling the denominator halves it.
2. **C.** After a long time, the capacitor voltage equals the battery voltage. The resistor voltage and circuit current are zero. During charging, charge moves in the wires onto and off the plates; it does not cross the ideal dielectric.
3. **D.** The sine of the angle is zero.
4. **B.** Each current experiences the magnetic field of the other wire.
5. **B.** Faraday law depends on the rate of flux change.
6. **D.** A plane electromagnetic wave in vacuum is transverse.

Part II: Written solutions

1. V = IR = 6 V. P = VI = 12 W (or I^2R). Two marks per result.
2. Total resistance = 12 Ω; current = 2 A; drops are 8 V and 16 V. One mark each; the drops must sum to 24 V.
3. F = BIL = 0.30 times 2 times 0.50 = 0.30 N. Four marks for formula, perpendicular condition, substitution and unit.
4. B is inversely proportional to r, so B = 30 times 2/6 = 10 microtesla. Credit model, ratio, arithmetic and unit.
5. Magnitude = 20 times 0.03/0.10 = 6 V. The direction needs the direction of the flux change; it cannot be inferred from these magnitudes alone.
6. RMS = 20/square root of 2 = 14.14 V. Average power = RMS squared/10 = 20 W. Two marks per stage.

Optics and Waves / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** Light is electromagnetic and does not require a material medium.
2. **A.** Wavelength equals speed divided by frequency.
3. **A.** The group is the envelope formed by combining nearby frequencies.
4. **C.** Beat frequency is the absolute difference.
5. **C.** Photon energy is proportional to frequency.
6. **D.** The boundary does not change the frequency of the incident oscillation.

Part II: Written solutions

1. Frequency = 50/10 = 5 Hz. Period = $1\frac{0}{5}0 = 0.20$ s. Two marks each.
2. Speed = 8 times 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each.
3. Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.
4. Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.
5. A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship.
6. Sine of critical angle = $\frac{1}{1}.5 = 0.6667$, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Optics and Waves / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The minus sign indicates inversion and magnitude 2 gives the size ratio.
2. **A.** The long-focus objective and short-focus eyepiece give a large angular-magnification ratio.
3. **D.** Fringe spacing is inversely proportional to slit separation.
4. **B.** Their axes differ by 90 degrees and $\cos^2 90^\circ$ is zero.
5. **C.** Lens power is inverse focal length in metres, expressed in dioptres.
6. **D.** A controlled phase delay changes the polarization state.

Part II: Written solutions

1. 1/di = $\frac{1}{1}$2 - 1/24 = 1/24, so di = 24 cm. m = -1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.
2. Initial magnitude = $10\frac{0}{5}$ = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.
3. Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = (0.0000005 times 2)/0.0005 = 0.002 m = 2 mm. Credit both conversions, calculation and result.
4. First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.
5. Convert to -0.50 m. Power=1/(-0.50)=-2 dioptres. Credit conversion, formula, sign and unit.
6. Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Calculus I / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The derivative is 2x, so its value at 3 is 6.
2. **D.** A constant function has zero rate of change.
3. **A.** The product rule differentiates each factor in turn.
4. **D.** The inner derivative contributes a factor of 5.
5. **C.** Implicit differentiation gives slope $-\frac{x}{y}$.
6. **B.** This is the standard inverse-tangent derivative.

Part II: Written solutions

1. [3(x+h) squared -3x^2]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.
2. f(2)=8; derivative=3x^2; slope at 2 is 12. Tangent y-8=12(x-2), equivalently y=12x-16. One mark per step.
3. Derivative = [(x-1)-(x+1)]/(x-1) squared = -2/(x-1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.
4. Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain -2x sin(x^2). Credit both layers, sign and product.
5. Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps.
6. Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x times (ln x+1). Four marks for these stages.

Calculus I / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** The rule concerns specified indeterminate quotient forms.
2. **A.** Negative slope means the function decreases as x increases.
3. **C.** The largest or smallest value may occur at either kind of candidate.
4. **B.** Equal endpoint heights make the average slope zero.
5. **D.** Differentiation loses constants, so all antiderivatives differ by a constant on an interval.
6. **C.** The odd integrand has equal and opposite signed areas.

Part II: Written solutions

1. The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.
2. Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,-4). One mark each for derivative, intervals and point.
3. Critical point x=1. Values: f(0)=0, f(1)=-1, f(3)=3. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.
4. The polynomial is continuous and differentiable. Average slope = (9-1)/(3-1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.
5. Set u=x^2+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.
6. Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4-1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Mechanics and Heat Lab / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** Slope units are vertical units divided by horizontal units.
2. **D.** The ideal mass is represented at the bob centre.
3. **D.** Rearranging h = gt^2/2 gives t^2 = (2/g)h.
4. **C.** Force = k times extension matches y = mx.
5. **C.** Period is proportional to the square root of mass.
6. **D.** Equal magnitude and opposite direction produce zero net force.

Part II: Written solutions

1. Percentage error = 100 times 0.2/9.8 = 2.04 percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained.
2. T=30/15=2 s. g=4π^2 times 0.99/4=9.77 m/s^2. Two marks per calculation.
3. g = 2/0.20 = 10 m/s^2. Four marks for axis identification, rearrangement, substitution and unit.
4. Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit.
5. Slope = 4π^2/k, so k=4π^2/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units.
6. Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction.

Mechanics and Heat Lab / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** Newton cooling uses the difference between object and surrounding temperatures.
2. **A.** Only pressure and volume are varied in the ideal Boyle experiment.
3. **A.** Expansion is proportional to original length.
4. **B.** Electrical power is VI; multiplying by time gives energy.
5. **B.** Energy flows from the hotter body to the cooler one until equilibrium.
6. **B.** The denominator determines the per-calorie unit.

Part II: Written solutions

1. Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation.
2. V2 = 120 times 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit.
3. Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 times 40)=0.000015 per K. Credit conversion, formula, substitution and result.
4. Energy=6 times 2 times 100=1200 J. Latent heat=120$\frac{0}{0}$.004=300000 J/kg. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter.
5. Water gains 0.10 times 4200 times 10=4200 J. Solid loses 0.20 times cs times 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result.
6. Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part.

Instruments and Lab Safety / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **D.** Risk depends on both the hazard and how people are exposed.
2. **C.** This prevents spreading contamination outside the work area.
3. **D.** Material can pass from the hands to food and the digestive system.
4. **A.** Oxidizers can support or intensify combustion.

Part II: Written solutions

1. Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action.
2. Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control.
3. Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation.
4. Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control.

Instruments and Lab Safety / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** The calibration mark defines the specified contained volume under its calibration conditions.
2. **B.** It is an engineering control for relevant airborne exposures.
3. **A.** Unknown hazards require assessment and the established response.
4. **B.** Administrative controls organize how work is authorized and performed.

Part II: Written solutions

1. Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices.
2. Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences.
3. It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention.
4. Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each.

Humans and the Environment / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** Temperature is a physical condition rather than a living organism.
2. **A.** They break down organic matter and return nutrients to the system.
3. **D.** Emigration removes individuals from the local population.
4. **B.** Mutualism is defined by benefits to both participants.

Part II: Written solutions

1. Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each.
2. Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency.
3. Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total.
4. Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation.

Humans and the Environment / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** Nutrient enrichment can drive excessive growth and subsequent oxygen depletion.
2. **C.** Demand exceeds the available biological capacity under the accounting model.
3. **D.** Avoiding waste reduces the need for later treatment and disposal.
4. **B.** The concept integrates all three dimensions and future consequences.

Part II: Written solutions

1. Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention.
2. It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation.
3. Prevention, reuse, recycling and disposal, respectively. One mark for each classification.
4. Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item.

Professional Development Seminar / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **D.** Professional practice combines expertise, preparation and ethical obligations.
2. **B.** Assessment information guides the next teaching decision.
3. **B.** Induction bridges initial preparation and independent professional practice.
4. **C.** Practice tests and develops professional knowledge in authentic settings.

Part II: Written solutions

1. Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair.
2. Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action.
3. Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases.
4. Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step.

Professional Development Seminar / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** A focused goal connects action to measurable change.
2. **C.** The reader can see what the evidence proves and how practice developed.
3. **D.** Evidence-based feedback supports improvement while respecting the colleague.
4. **A.** Pedagogical purpose should guide selection.

Part II: Written solutions

1. Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component.
2. A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation.
3. Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language.
4. Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation.